

Particle model of matter

3.1 Density



Queen
Elizabeth
School

Retrieval questions, answers on the whiteboards please!

1. What was it that Archimedes shouted out whilst running through the street?
2. What is the formula for density?
3. What is the scientific name for the Eureka can?
4. How do we work out the volume of a regular solid?

1. What happens to the molecules in a solid as it gets hotter....

- a) Start vibrating
- b) Vibrate faster
- c) Slow down

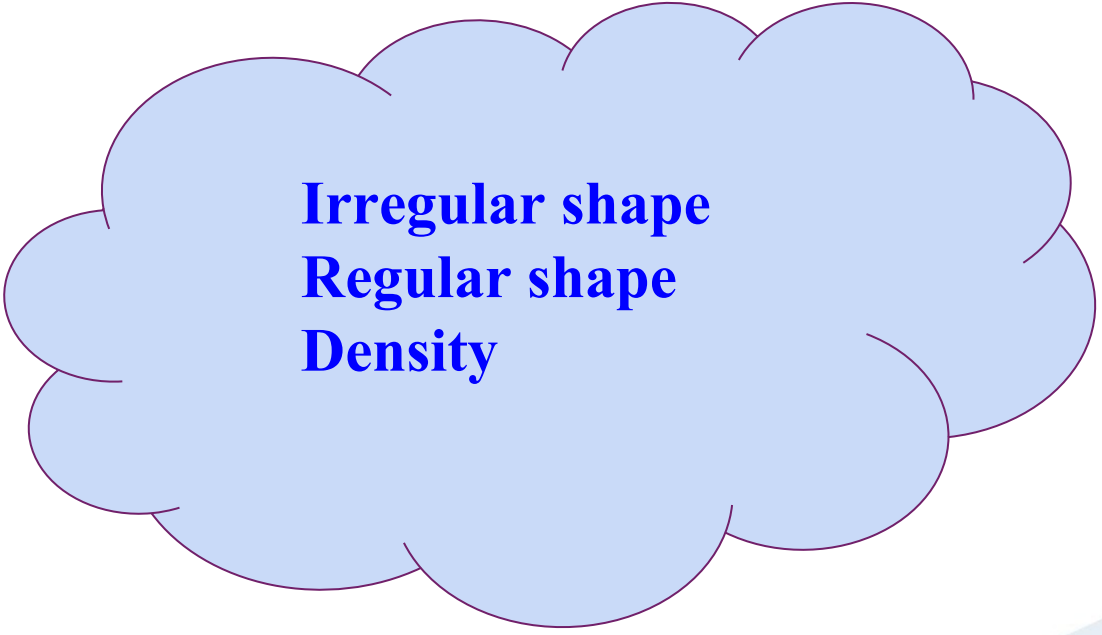
B

2. What happens to the molecules in a liquid as it gets hotter....

- a) Speed up
- b) Slow down
- c) Stay the same

a

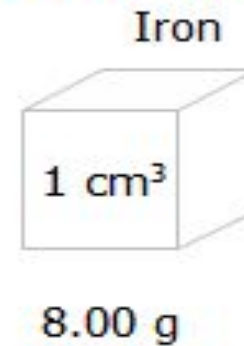
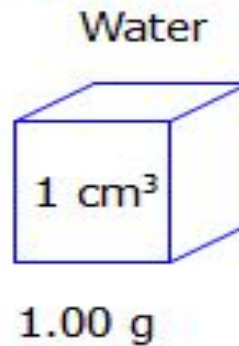
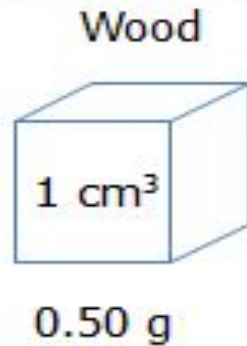
Understand density and how to measure and calculate density of regular and irregular shapes



Irregular shape
Regular shape
Density

Connector - What is Density?

If you take the same volume of different substances, then they will weigh different amounts.



IRON

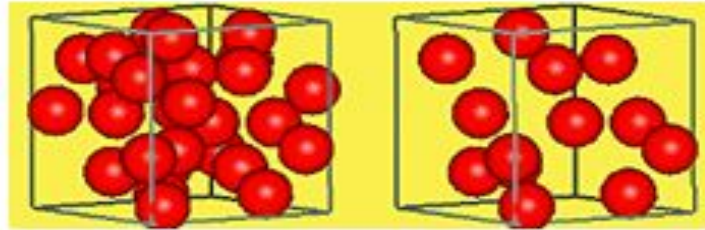
Q) Which has the greatest mass and therefore the most dense?

Density is the Mass per unit Volume

Density

Take a look at the two boxes below. Each box has the same volume.

If each ball has the same mass, which box would weigh more? Why?



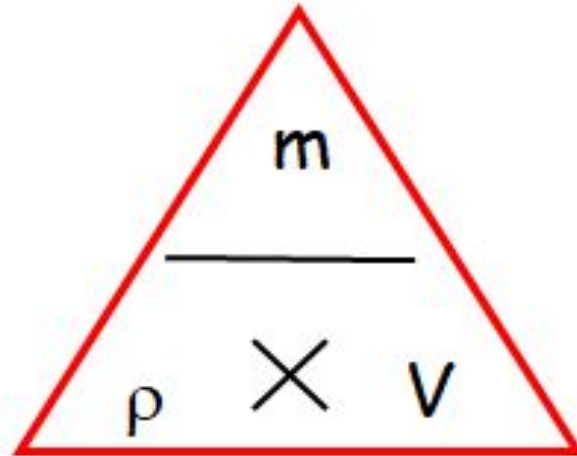
The box that has more balls has more mass per unit of volume.

This property of matter is called DENSITY.

Density Equation: You must be able to use and rearrange

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

g/cm³ or kg/m³ g or kg cm³ or m³



Let's do some practice together

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

If the Mass is 45g, and the volume is 5cm³, what's the density

45

5 = 9

practice

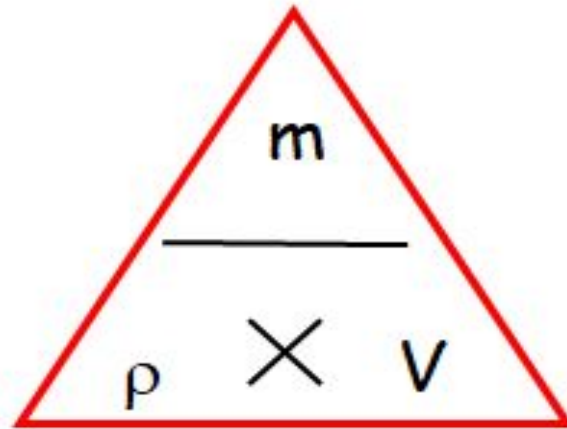
1. What is the density of a material if 10 cm^3 of it has a mass of 20 g ?
2. What is the density of a rock if it has a mass of $1,500 \text{ kg}$ and a volume of 0.5 m^3 ?

Let's now rearrange the equation

Density Equation: You must be able to use and rearrange

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

g/cm³ or kg/m³ g or kg cm³ or m³



Let's now rearrange the equation

Density Equation: You must be able to use and rearrange

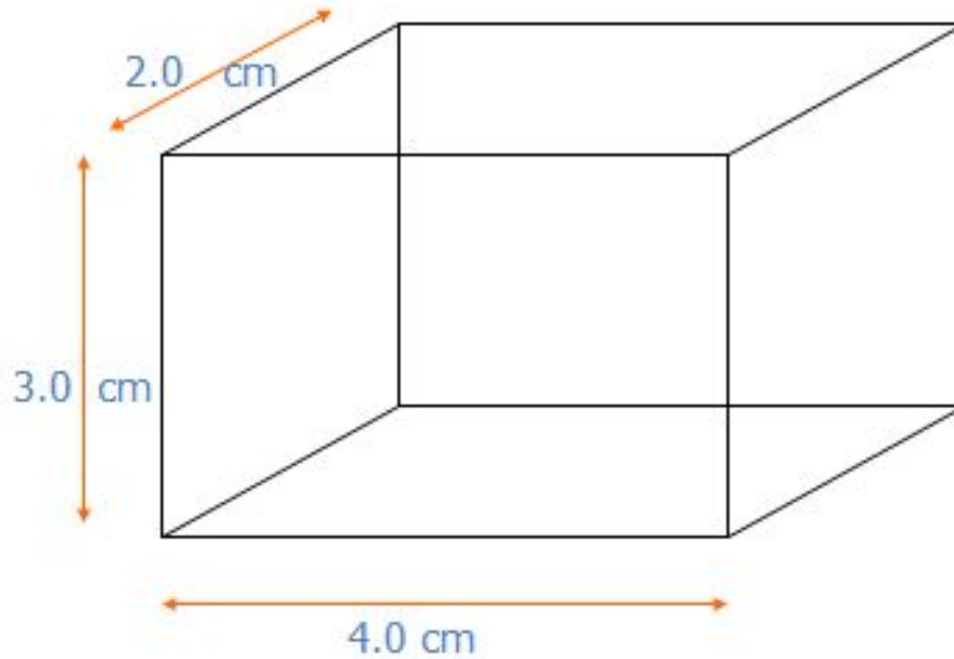
$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

g/cm³ or kg/m³ g or kg cm³ or m³

m

$\rho \times v$

How do you measure the volume of a REGULAR SOLID



- Measure the three lengths and calculate the Volume.
(i.e. Volume = length x width x height)

$$4 \times 2 \times 3$$



What is density.....

- a) Mass per unit volume
- b) Volume per unit mass
- c) Force per unit area

a

How do you calculate density.....

a) Mass $\div$ volume

b) Distance $\div$ time

c) Volume $\div$ mass

a

What are the units of density....

a) N/m^2

b) kg/m^2

c) kg/m^3

C

Calculating density practice worksheet. Please do this on the worksheet so you can stick in your books. Remember no folding the sheet. Complete in pencil until you are 100% sure the answers are correct.

An irregular solid could be something like this stone. How would we be able to work out its density?

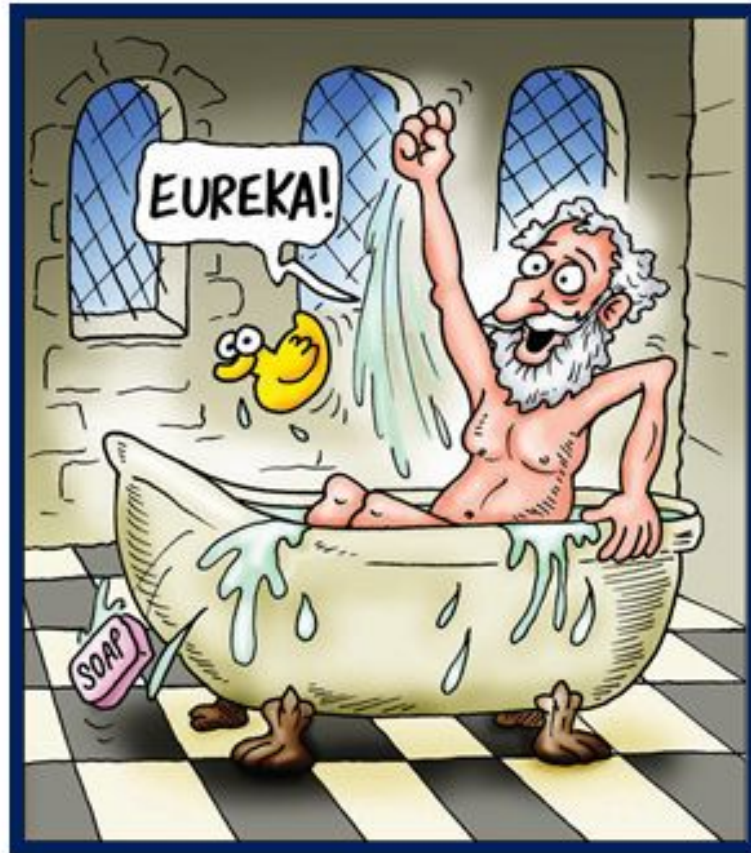


Archimedes' Principle

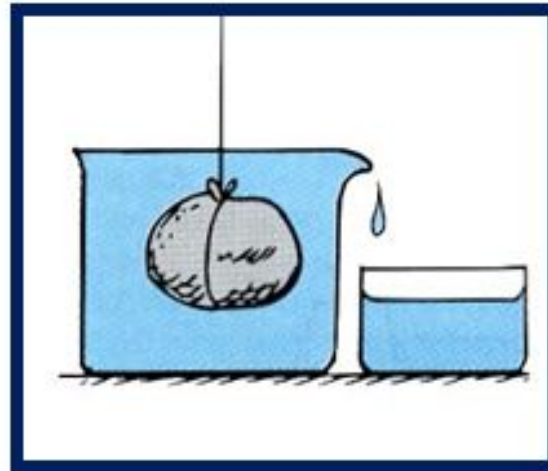
- Does anyone know the story of Archimedes' bath?
- Why did he shout 'Eureka!'?

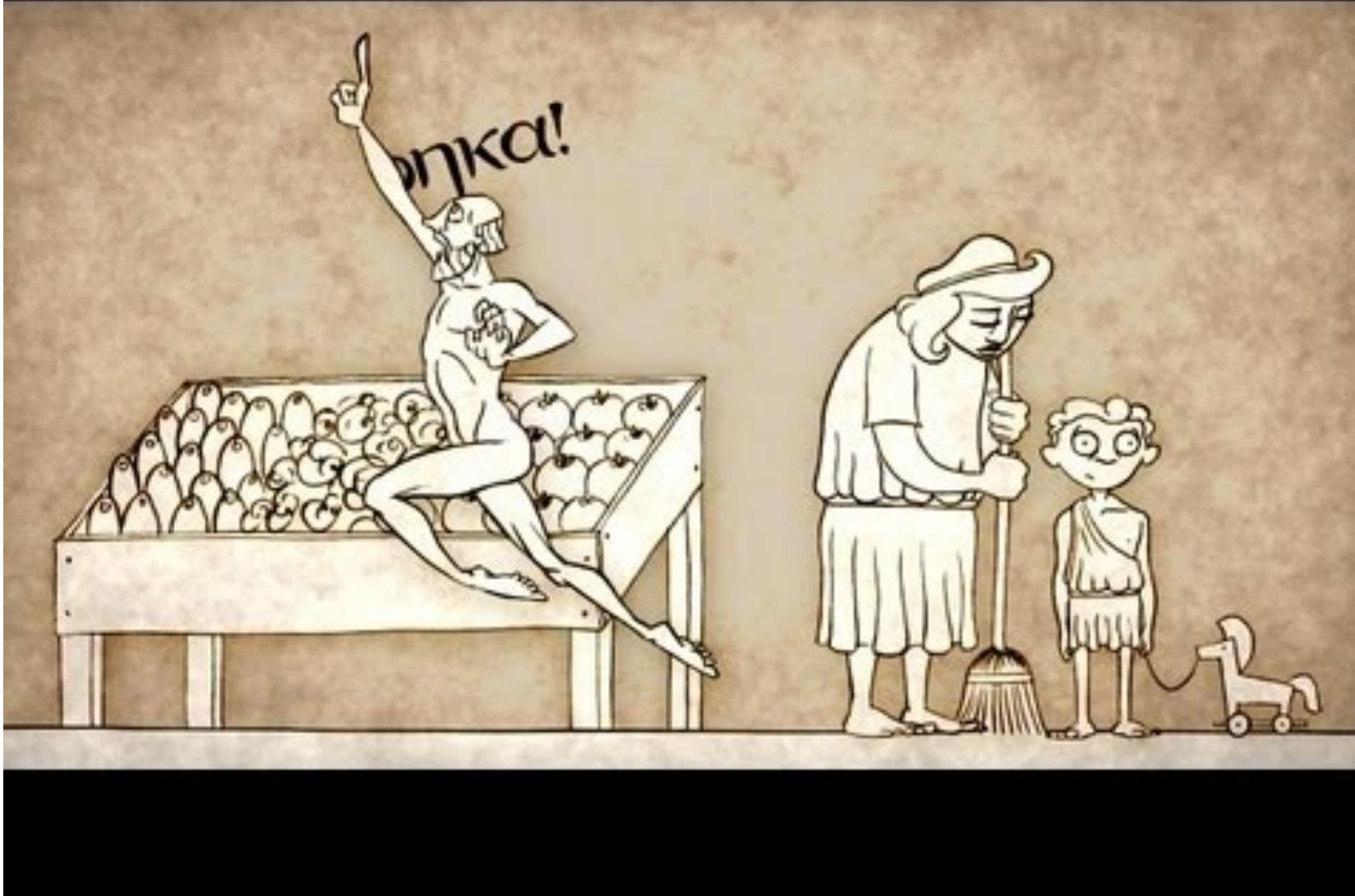


Archimedes' Principle



He found that the volume of water displaced is equal to the volume of the object





Calculating density practice worksheet. Please do this on the worksheet so you can stick in your books.

Complete in pencil until you are 100% sure the answers are correct.